

Local decisions. Shared safety layer. Human oversight.

A coordination prototype for crowded orbit · YOHAKU 2026 Challenge 7

The problem

Thousands of satellites from different operators share orbit with debris, and nothing decides who moves. In 2019 ESA's Aeolus dodged Starlink 44 alone after a follow-up email went unseen. As AI takes over these decisions, priority, fairness and responsibility remain open.

The system

Layer	What it does
Onboard AI	Every satellite runs the same small policy (multi-agent RL). It senses its most dangerous neighbour and proposes when and how to burn.
Handshake	PROPOSE / ACK / DO-NOT-MOVE / EXECUTED / ESCALATE replaces email. A silent party is treated as unable to move.
Safety layer: priority check	Both onboard AIs compute the same priority from shared data: debris never moves; crewed holds; low fuel protected; free-riders yield; autonomous yields; commercial yields to public-good; then fair tie-breaks.
Safety layer: hard triggers	Crewed, high risk, a safety-layer conflict or low confidence go to a human; the AI cannot switch them off. If the AI proposes something the safety layer forbids, the safe action is executed and the conflict is logged.
Fairness ledger	Double-entry: whoever yields earns credit; debris owners, silence and false declarations pay; balances sum to zero.

Results (10,000 held-out scenarios from real ESA conjunction data, illustrative physics)

Strategy	Collisions	Mean Δv (m/s)	Burden Gini
Do nothing	7.91 %	0	0
Both burn when dangerous	0.34 %	0.053	0.20
Lower ID yields	1.04 %	0.030	0.54
Shared onboard AI (with safety layer)	0.66 %	0.044	0.43

The onboard AI ended 99.34 % of scenarios safely (95 % CI 99.16-99.48) and resolved 91.7 % of the encounters that would otherwise collide, always inside the safety layer. The same network runs live in the browser.

Governance

Question	Answer
Who is responsible?	Every failure mode has an owner: data provider for bad data, AI developer within its declared domain, operator for silence, false declarations and overrides, licensing state for supervision (OST Art. VI).
Who decides what the network learns?	Nobody alone: cases are logged, updates are proposed, an independent Orbital Incident Board approves, versions are recorded. Hard triggers are outside learning.
What does each actor owe? (NARETU)	Operators: data, a timely handshake, a fair share of dodges, disposal. AI developers: explainability, a declared domain, unswitchable triggers. Data providers: neutral warnings. States: supervision and verified registration. Everyone: a usable orbit for those not yet launched.

Inspiration and its limits

From the challenge's mycelium lens we borrowed local sensing, a connected network and rewards for partners that give back. We did not borrow its morality: fungi cannot be held responsible and repair themselves; orbit cannot. NARETU framing credited to Chief Titus Letaapo and Samburu obligations-based governance. Data: CelesTrak (T.S. Kelso); ESA Kelvins dataset (CC-BY-4.0).